

Where Does the Investment Demand Shock Come From?

A toy model of idiosyncratic shocks and lumpy investment

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Building on *Nirei and Ragot (2025)*

What this toy model is for

A micro-foundation that answers three questions for the estimation:

1. What do we mean by **propagation of an idiosyncratic shock**?
2. Why is the **lumpy-investment** assumption essential?
3. Why is it legitimate to use a **reduced-form investment demand shock** (a shock added directly on investment) in estimation?

Logic of the argument:

- ▶ A small i.i.d. shock pushes a few firms over their investment threshold.
- ▶ In general equilibrium this raises output, which moves every firm's threshold.
- ▶ The induced wave does **not** die out — it is self-sustaining and, in the limit of many firms, survives as a **degenerate (non-random) aggregate** investment shock.

Intermediate-goods producers

Standard New-Keynesian CES aggregator and demand:

$$Y_t = \left(\int_0^1 y_{jt}^{\frac{\eta-1}{\eta}} dj \right)^{\frac{\eta}{\eta-1}}, \quad y_{jt} = \left(\frac{p_{jt}}{P_t} \right)^{-\eta} Y_t, \quad y_{jt} = a_{jt} k_{jt}.$$

Linear-in-capital technology $y_{jt} = a_{jt} k_{jt}$. Combining demand and production, real revenue is

$$r_{jt}(a_{jt}, k_{jt}) = \frac{p_{jt} y_{jt}}{P_t} = (a_{jt} k_{jt})^{1-\frac{1}{\eta}} Y_t^{\frac{1}{\eta}}.$$

- ▶ Concave in own capital ($\eta > 1$): diminishing returns through the demand curve.
- ▶ Revenue, hence the desired capital, scales with aggregate output Y_t — the channel through which one firm's action will later spill over to all firms.

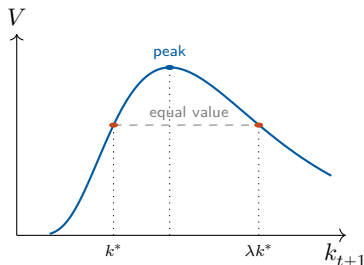
Lumpy investment and the investment threshold

Firms solve

$$\max_{\{x_{t+k}\}} \mathbb{E}_t \sum_{k \geq 0} \Lambda_{t,t+k} (r_{j,t+k} - x_{j,t+k}),$$

$$\text{s.t. } k_{j,t+1} = (1 - \delta)k_{j,t} + x_{j,t},$$

$$x_{j,t+k} \in \{0, \lambda k_{j,t+k}\}.$$



Indifference at k^* (the firm knows its next-period technology) pins down the threshold:

$$k_{j,t+1}^* = a_{j,t+1}^{\eta-1} \Phi_t(\Lambda_t, \Lambda_{t+1}) Y_{t+1}$$

- Desired capital scales with aggregate output Y_{t+1} — the spillover channel that will carry one firm's action to all firms.

Normalize a firm's position by its **distance to the threshold**

$$s_{jt} = \frac{\log(k_{jt}/k_{jt}^*)}{\log \lambda},$$

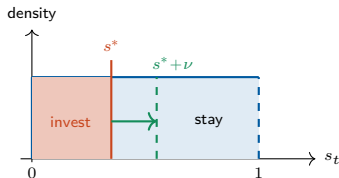
so that investing means jumping up exactly one notch ($s \mapsto s + 1$) in these units.

Cross-section stationary distribution

- ▶ Assume $a_t \in \{a_L, a_H\}$ and $s_t \mid a_t \sim U[0, 1)$ under stationary distribution.
- ▶ Macro variables are constant: $Y = Y_t = Y_{t+1}$, $\Phi = \Phi_t = \Phi_{t+1}$

Without investing, position drifts by

$$s_{j,t+1} = s_{j,t}(a) - \underbrace{\frac{-\log(1-\delta) + (\eta-1) \log \frac{a'}{a}}{\log \lambda}}_{s^*(a, a')}.$$



- ▶ Since $s_t \sim U[0, 1)$, $s^*(a, a')$ is the **investing fraction**
- ▶ Add an i.i.d. shock ε on a' *only* in group $(a, a') = (b, b')$:

$$s^*(b, \varepsilon b') = s^*(b, b') + \underbrace{\frac{(\eta-1) \log \varepsilon}{\log \lambda}}_{\nu}.$$

- ▶ The cutoff in that group rises by ν , so an extra ν -mass invests.

An idiosyncratic shock in one group

$a = H$	$\omega(H, L)$	$\omega(b, b')$
$a = L$	$\omega(L, L)$	$\omega(L, H)$
	$a' = L$	$a' = H$

Figure 3. Four groups (a, a') with masses ω ;
the shock hits group (b, b') .

More investment in (b, b') raises *future aggregate output*:

$$Y_{t+1}(b, b', \nu) = \left(\omega(b, b') \int_0^1 (b' k_{t+1}(s_t))^{\frac{\eta-1}{\eta}} dF(s_t | b) \right)^{\frac{\eta}{\eta-1}},$$

split at the cutoff $s^*(b, b') + \nu$ into investing firms and the rest. This output increase is **propagation**.

Propagation through the threshold, and why it does not die out

Higher Y_{t+1} raises $k^* \propto Y_{t+1}$ everywhere, shifting the cutoff in *every* group:

$$s_t^*(a, a', \nu) = \frac{-\log(1 - \delta) + (\eta - 1) \log(a'/a) + \log Y_{t+1}(\nu) - \log Y_t}{\log \lambda}.$$

Per-group and aggregate ν -induced fraction:

$$\Delta s_t^*(a, a', \nu) = \frac{\log Y_{t+1}(\nu) - \log Y_t}{\log \lambda}, \quad \Delta s_t^*(\nu) = \sum_{(a, a')} \omega(a, a') \Delta s_t^*(a, a', \nu).$$

- ▶ The first ν of investing firms raises Y , which induces $\Delta s^*(\nu)$ more, which raises Y again ... a GE multiplier that iterates to a new equilibrium.
- ▶ Key theoretical property (small-shock limit):

$$\lim_{\nu \rightarrow 0} \frac{\Delta s_t^*(\nu)}{\omega(b, b') \nu} = 1$$

the ν -induced fraction has the *same scale* as the original ε -induced fraction $\Rightarrow$ the wave does **not** attenuate across iterations.

Why a reduced-form macro investment demand shock is justified

The tension. At each iteration more firms invest, but each adds only a small amount; if the per-firm amount decayed fast enough, aggregate investment would barely move.

$$\text{Total investment} = \sum_{i=1}^{\infty} \text{induced investing fraction}^i \times \text{average investment}^i$$

Where lumpiness bites. With lumpy investment, “if you invest, you must invest a large amount” ($x = \lambda k$). The fraction result above ($\lim = 1$) keeps the *mass* of investors from shrinking, and lumpiness keeps each contribution *large*.

Theoretical result

As the number of firms $\rightarrow \infty$, the i.i.d. idiosyncratic shock ε vanishes with **asymptotic variance to zero**, yet the induced aggregate investment converges to a **non-degenerate random variable**.

Implication for estimation. The micro idiosyncratic shock aggregates into a well-defined, non-vanishing movement in **investment**. So it is legitimate to model it as a reduced-form shock placed **directly on investment**, rather than on technology.

Takeaways

1. **Propagation** = a within-group idiosyncratic shock raising aggregate output, which feeds back onto every firm's investment threshold.
2. **Lumpy investment** is essential: it creates the threshold (the investing fraction s^*) and guarantees each induced investment is large, so the aggregate effect does not wash out.
3. **Reduced-form ID shock** is justified: the aggregated micro shock survives as a degenerate shock on investment, validating adding it directly to investment in estimation.